

Elias Ramzi | AI Research Scientist

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Researcher specialized in Deep Learning and Computer Vision.

Experiences

valeo.ai (Paris, France) - full-time

2024-

AI research scientist

Deep learning research interests: world models, vision-language models, LLMs.

- Working on world models (simulation & planning) for autonomous driving.
- Supervising a PhD student with focus on LLMs & VLMs and explainability.
- Scaling model training, with models up to 1.9B parameters, trained on 192 H100s. Training conducted with μP and deepspeed2.
- Publication of research papers, and technical report.

Cnam & Coexya (Paris, France) - 3 year PhD

2021-2024

PhD student

My PhD in deep learning was focused on computer vision and image retrieval:

- Worked in computer vision for image retrieval, on topics such as ranking losses and hierarchical data from both mathematical and experimental aspects.
- Published four papers in major international machine learning conferences (NeurIPS, ECCV $\times$ 2, ICML), three of which as first author, and 2 papers in journals (TPAMI, TMLR). Adapted a research project (HAPPIER, ECCV 2022) to trademark logo retrieval, HAPPIER is now used to train models that go in production in Coexya's Accepto.
- Used the public HPC cluster Jean Zay ☞ to train neural networks in a distributed fashion with up to 4 nodes totaling 16 GPUs.
- In June 2025, my PhD was awarded with the "Prix de Thèse" of AFRIF.

Cnam (Paris, France) - 5 month internship

2020

Research intern

Deep learning for 3D medical image segmentation. I used a learned confidence measure (ConfidNet) to combine segmentation results from different 2D U-Nets.

Sancare (Paris, France) - 6 month internship

2019

Data engineer and data scientist intern

Balto (Sydney, Australia) - 6 month internship

2018

Data engineer intern

Education

PhD in Computer Science, Cnam & Coexya (France)

2021-2024

I investigate deep learning approaches to image retrieval. Specifically, I work on designing appropriate ranking losses to train deep neural networks for image retrieval. I also contributed on out-of-distribution detection using energy-based models, worked on collaborative filtering recommendation using graph neural networks and on VLM adaptation with prompt learning.

Supervisor: Nicolas Thome ☞ (Sorbonne Université, Paris, France).

Advisors: Nicolas Audebert ☞ (IGN, Paris, France); Clément Rambour ☞ (Sorbonne Université, Paris, France).

Industrial advisor: Xavier Bitot ☞ (Coexya, Paris, France).

Engineering diploma, CentraleSupélec (France)

2016-2020

Courses in signal processing (sound, image, speech), machine learning, statistical models (estimators, Bayesian learning), data science (data mining, sparse data representation, NLP).

Technical skills

Open source projects

- Code to reproduce VaViM and VaVAM:  VideoActionModel
- Code to train global and local prompts for vision-language models (ECCV'24):  GalLoP
- Code to train out-of-distribution detectors with HEAT (ICML'23):  HEAT.
- First of its kind hierarchical landmark dataset (TPAMI, under review):  google-landmark &  SupRank.
- Code to train models with HAPPIER (ECCV'22) for hierarchical image retrieval:  HAPPIER.
- Code to train models with ROADMAP (NeurIPS'21) for image retrieval:  ROADMAP.

Technologies

- My programming language: Python (PyTorch, Lightning, NumPy, Hydra, Pandas, Scikit-learn).
- My work environment : Linux/macOS + Git + VS Code & GitHub Copilot.
- Other tools: L^AT_EX, Shell script, SLURM, HyperQueue, Nextflow.

Contributions to open-source

- Pytorch metric learning:  pytorch-metric-learning.
- Implementation of Mixture of Experts:  mixture-of-experts.
- Pytorch-lightning:  lightning.
- Universal Image Embeddings:  Universal-Image-Embeddings.

Miscellaneous

- French is my mother tongue and I speak English at C1 level.
- I enjoy (trail) running, biking (and sometimes swimming) and participating in organized races. I also like hiking and climbing.

Publications

Florent Bartoccioni, **Elias Ramzi**, Victor Besnier, Shashanka Venkataramanan, Tuan-Hung Vu, Yihong Xu, Loick Chambon, Spyros Gidaris, Serkan Odabas, David Hurych, Renaud Marlet, Alexandre Boulch, Mickael Chen, Éloi Zablocki, Andrei Bursuc, Eduardo Valle, Matthieu Cord. "VaViM and VaVAM: Autonomous Driving through Video Generative Modeling" *Technical report*, 2025. online: arxiv.org/abs/2502.15672

Amaia Cardiel, Éloi Zablocki, **Elias Ramzi**, Oriane Siméoni, and Matthieu Cord. "LLM-wrapper: Black-Box Semantic Aware Adaptation of Vision-Language Models for Referring Expression Comprehension." *ICLR*, 2025. online: arxiv.org/abs/2409.11919

Elias Ramzi, Nicolas Audebert, Clément Rambour, André Araujo, Xavier Bitot and Nicolas Thome. "Optimization of Rank Losses for Image Retrieval." *TPAMI*, 2025. online: arxiv.org/abs/2309.08250

Marc Lafon, **Elias Ramzi**, Clément Rambour, Nicolas Audebert, Nicolas Thome. "GalLoP: Learning Global and Local Prompts for Vision-Language Models." *ECCV*, 2024. online: arxiv.org/abs/2407.01400

Yannis Karmim, **Elias Ramzi**, Raphaël Fournier S'niehotta, and Nicolas Thome. "ITEM: Improving Training and Evaluation of Message-Passing based GNNs for top-k recommendation." *TMLR*, 2024. online: arxiv.org/abs/2407.07912

Marc Lafon, **Elias Ramzi**, Clément Rambour, Nicolas Thome. "Hybrid Energy Based Model in the Feature Space for Out-of-Distribution Detection." *ICML*, 2023. online: arxiv.org/abs/2305.16966

Elias Ramzi, Nicolas Audebert, Nicolas Thome, Clément Rambour, and Xavier Bitot. "Hierarchical Average Precision Training for Pertinent Image Retrieval." *ECCV*, 2022. online: arxiv.org/abs/2207.04873

Elias Ramzi, Nicolas Thome, Clément Rambour, Nicolas Audebert, and Xavier Bitot. "Robust and Decomposable Average Precision for Image Retrieval." *NeurIPS*, 2021. online: arxiv.org/abs/2110.01445